

Hierarchical models and genetic heterogeneity

Lars Rönnegård

Dalarna University

Swedish University of Agricultural Sciences

Presentation at the Roslin Institute

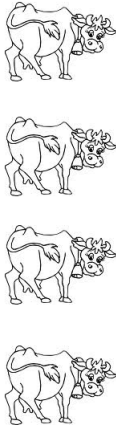


Outline

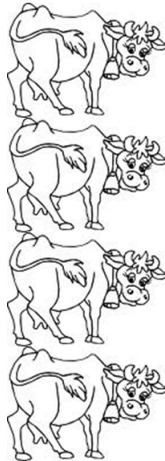
- Using Generalized Linear Models for
 - Finding genes for uniformity and phenotypic robustness
 - The hglm package in R
 - GWAS with shrinkage smoothing
 - Modelling phenotypic robustness in animal breeding

Ordinary (mean-controlling) genes

SNP Type 1



SNP Type 2



Variance-controlling genes

SNP Type 1



SNP Type 2



Results from Arabidopsis

Shen, Pettersson, Rönnegård and Carlborg (2012). *Inheritance beyond plain heritability: variance controlling genes in Arabidopsis thaliana*. PLoS Genetics 8(8):e1002839.

- Arabidopsis thaliana study including 199 individuals
- Trait: molybdenum content
- 216,130 SNP

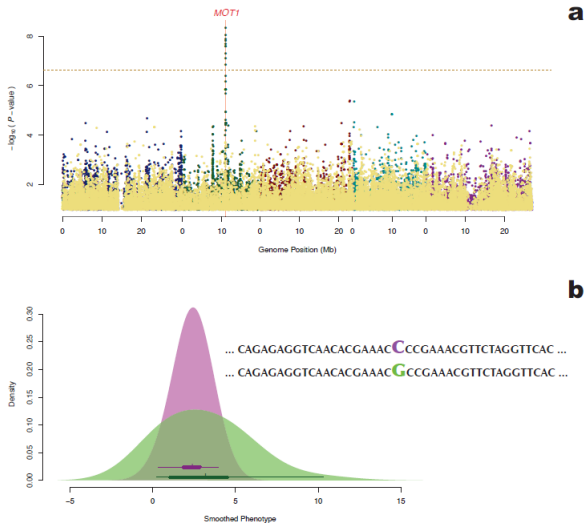


Figure 2: A gene controlling robustness of molybdenum contents in Arabidopsis. Top figure (a) shows logP values for mean-controlling SNP (yellow) and variance-controlling SNP (different colours for different chromosomes). Bottom figure (b) shows substitution effect of the *MOT1* allele. (Shen *et al.* 2012 PLoS Genetics)

Levene's test for inequality of variance

We have observations Y with sample size N divided into k subgroups.
To compute Levene's test statistic we just need

$$Z_{ij} = |Y_{ij} - \bar{Y}_{i.}|$$

with Y_{ij} the value of Y for the j :th observation of the i :th subgroup, and with bars indicating subgroup means.

The test statistic is $W = \frac{(N-k) \sum_{i=1}^k N_i (\bar{Z}_{i.} - \bar{Z}_{..})^2}{(k-1) \sum_{i=1}^k \sum_{j=1}^{N_i} (Z_{ij} - \bar{Z}_{i.})^2}$

The test rejects the hypothesis that the variances are equal if:

$$W > F_{(\alpha, k-1, N-k)}$$

Pros: Not sensitive to deviations from normality

Cons: Not possible to include covariates and random effects (to adjust for family structure).

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Detecting Major Genetic Loci Controlling Phenotypic Variability in Experimental Crosses

Lars Rönnegård^{*:1} and William Valdar[†]

^{}Statistics Unit, Dalarna University, SE-781 70 Borlänge, Sweden and [†]Department of Genetics, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina 27599-7265*

Manuscript received January 20, 2011

Accepted for publication March 21, 2011

ABSTRACT

Traditional methods for detecting genes that affect complex diseases in humans or animal models, milk production in livestock, or other traits of interest, have asked whether variation in genotype produces a change in that trait's average value. But focusing on differences in the mean ignores differences in variability about that mean. The robustness, or uniformity, of an individual's character is not only of great practical importance in medical genetics and food production but is also of scientific and evolutionary interest (*e.g.*, blood pressure in animal models of heart disease, litter size in pigs, flowering time in plants). We describe a method for detecting major genes controlling the phenotypic variance, referring to these as vQTL. Our method uses a double generalized linear model with linear predictors based on probabilities of line origin. We evaluate our method on simulated F_2 and collaborative cross data, and on a real F_2 intercross, demonstrating its accuracy and robustness to the presence of ordinary mean-controlling QTL. We also illustrate the connection between vQTL and QTL involved in epistasis, explaining how these concepts overlap. Our method can be applied to a wide range of commonly used experimental crosses and may be extended to genetic association more generally.

Model

Traditional model for single SNP regression

$$\mathbf{y} = \mu + \mathbf{x}\mathbf{b} + \mathbf{e}$$

$$\mathbf{e} \sim N(0, \sigma_e^2)$$

Model to detect variance-controlling genes

$$\mathbf{y} = \mu + \mathbf{x}\mathbf{b} + \mathbf{e}$$

$$e_i \sim N(0, \sigma_{e,i}^2) ; \log(\sigma_e^2) = c + xv$$

can also be expressed as

$$\mathbf{y} \sim N(\mu + \mathbf{x}\mathbf{b}, \exp(c + xv))$$

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Using Generalized Linear Models

Appl. Statist. (1987)
36, No. 3, pp. 332–339

Modelling Variance Heterogeneity in Normal Regression Using *GLIM*

By MURRAY AITKIN†

University of Lancaster, UK

[Received August 1985. Revised January 1987]

SUMMARY

This paper describes and presents simple *GLIM* macros for the modelling of variance heterogeneity in normal regression analysis, using a log-linear regression model for the variance. The procedure is illustrated with two examples.

Keywords: Variance heterogeneity; Score test; GLIM; maximum likelihood

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Traditional model for SNP regression

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model1 <- glm(y ~ SNP, family = gaussian(link = identity))
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Model easy to fit using Gordon K. Smyth's dglm package in R

```
library(dglm)
```

```
model2 <- dglm(y ~ SNP, ~ SNP, family = gaussian(link = identity))
```

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R package hglm

20

CONTRIBUTED RESEARCH ARTICLES

hglm: A Package for Fitting Hierarchical Generalized Linear Models

by Lars Rönnegård, Xia Shen and Moudud Alam

Abstract We present the **hglm** package for fitting hierarchical generalized linear models. It can be used for linear mixed models and generalized linear mixed models with random effects for a variety of links and a variety of distributions for both the outcomes and the random effects. Fixed effects can also be fitted in the dispersion part of the model.

Introduction

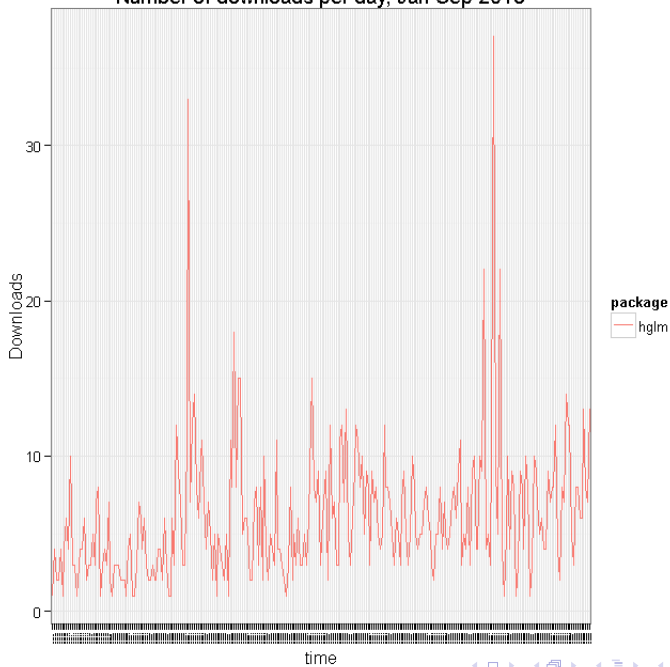
The **hglm** package (Alam et al., 2010) implements the estimation algorithm for hierarchical generalized linear models (HGLM; Lee and Nelder, 1996).

effects, respectively, in the linear predictor for the means and X_{disp} is the design matrix for the fixed effects in the dispersion parameter. This enables a more flexible modeling of the random effects than specifying the model by an R formula. Consequently, this option is not as user friendly but gives the user the possibility to fit random regression models and random effects with known correlation structure.

The **hglm** package produces estimates of fixed effects, random effects and variance components as well as their standard errors. In the output it also produces diagnostics such as deviance components and leverages.

Three illustrating models

"Number of downloads per day, Jan-Sep 2013"



The hglm package

Fitting variance-controlling gene effects and random effects:

```
library(hglm)
model2 <- hglm(fixed = y ~ SNP,
               disp = ~ SNP,
               random = ~ 1|ID,
               family = gaussian(link = identity) )
```

Other possibilities in hglm

Notation using design matrices:

```
model2 <- hglm(X, y, Z, X.disp, family = gaussian(link = identity) )
```

Possible to fit animal model, random regression, etc. For instance:

```
animal.model <- hglm(X, y, Z = t(chol(A)),  
                     family = gaussian(link = identity) )
```

Possible to fit several random effects:

```
model3 <- hglm(X, y, Z = cbind(Z1,Z2), randC = c(ncol(Z1),ncol(Z2)),  
              family = gaussian(link = identity) )
```

Possible to fit other distributions:

```
model4 <- hglm(X, y, Z, family = poisson(link = log) )
```

Possible to fit other distributions for the random effects too:

```
negative_binomial.model <- hglm(X, y, Z, family = poisson(link = log),  
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A DHGLM for GWAS and genomic prediction

A two-level model with shrinkage smoothing

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \mathbf{e}$$

$$\mathbf{e} \sim N(0, \sigma_e^2)$$

$$\mathbf{u}_i \sim N(\mathbf{0}, \sigma_{u,i}^2), \log(\sigma_u^2) = c + \mathbf{Z}\mathbf{v}$$

$$\mathbf{v} \sim N(0, \mathbf{V}\sigma_d^2)$$

where $\mathbf{V}$ specifies a correlation matrix so that the correlation between $\mathbf{v}_k$ and $\mathbf{v}_l$ is equal to $r^{|k-l|}$ and r is chosen to give reasonable smoothing.

- Rönnegård and Lee (2010) "Hierarchical generalized linear models have a great potential in genetics and animal breeding" In proceedings: World Congress on Genetics Applied to Livestock Production, Leipzig, Germany
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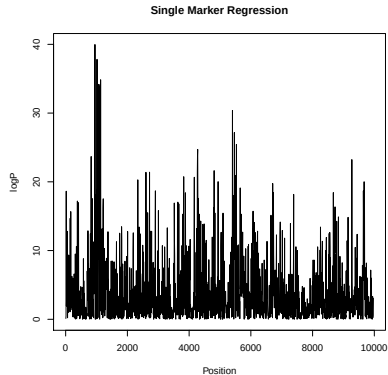
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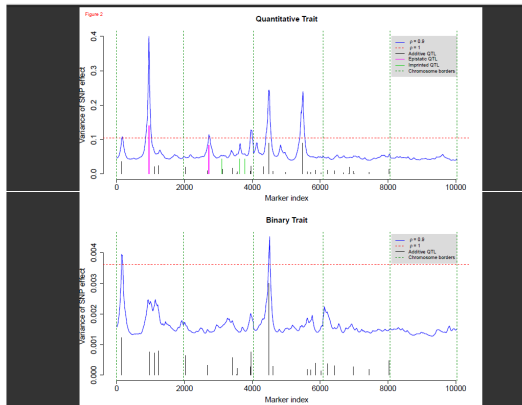
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QTLMAS 2010 Workshop simulated data

- 2000 individuals and 10,000 SNPs on five chromosomes
- Two traits: normal and binary



A DHGLM GWAS



Heteroscedastic Effects Model: an approximation without smoothing

INVESTIGATION

A Novel Generalized Ridge Regression Method for Quantitative Genetics

Xia Shen,^{*,†,‡} Moudud Alam,[†] Freddy Fikse,[†] and Lars Rönnegård^{†,‡}

^{*}Division of Computational Genetics, Department of Clinical Sciences and [†]Department of Animal Breeding & Genetics, Swedish University of Agricultural Sciences, 75007 Uppsala, Sweden, and [‡]Statistics, School of Technology and Business Studies, Dalarna University, 78170 Borlänge, Sweden

ABSTRACT As the molecular marker density grows, there is a strong need in both genome-wide association studies and genomic selection to fit models with a large number of parameters. Here we present a computationally efficient generalized ridge regression (RR) algorithm for situations in which the number of parameters largely exceeds the number of observations. The computationally demanding parts of the method depend mainly on the number of observations and not the number of parameters. The algorithm was implemented in the R package **bigRR** based on the previously developed package **hglm**. Using such an approach, a *heteroscedastic effects model* (HEM) was also developed, implemented, and tested. The efficiency for different data sizes were evaluated via simulation. The method was tested for a bacteria-hypersensitive trait in a publicly available *Arabidopsis* data set including 84 inbred lines and 216,130 SNPs. The computation of all the SNP effects required <10 sec using a single 2.7-GHz core. The advantage in run time makes permutation test feasible for such a whole-genome model, so that a genome-wide significance threshold can be obtained. HEM was found to be more robust than ordinary RR (a.k.a. SNP-best linear unbiased prediction) in terms of QTL mapping, because SNP-specific shrinkage was applied instead of a common shrinkage. The proposed algorithm was also assessed for genomic evaluation and was shown to give better predictions than ordinary RR.

Arabidopsis data

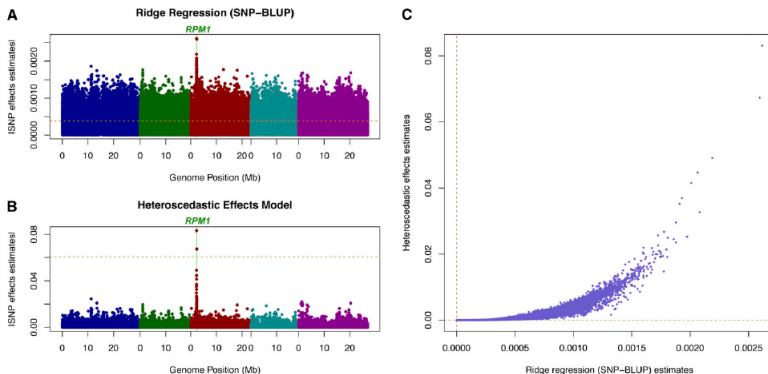


Figure 2 Estimated SNP effects for the *Arabidopsis* bacteria-hypersensitive trait AvrRpm1 (Atwell *et al.* 2010) using (A) ridge regression (SNP-BLUP) and (B) heteroscedastic effects model, which are plotted against each other in C. The horizontal dashed lines in A and B indicate the 5% genome-wide significance threshold from a randomization test using 1000 permutations.

Implemented in bigRR

```
library(bigRR)
data(Arabidopsis) #Contains y and Z
X <- model.matrix(y~1)
SNPBLUP.result <- bigRR(y = y, X = X, Z = scale(Z),
                        family = binomial(link = 'logit'))
HEM.result <- bigRR_update(SNPBLUP.result, scale(Z),
                           family = binomial(link = 'logit'))
```

Some details to note

- Computations <50 s on my old computer
- Transforms both effects and PEV from GBLUP to SNP-BLUP

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Inherited phenotypic robustness

- Rönnegård, Felleki, Fikse, Mulder and Strandberg (2010) *Genetic heterogeneity of residual variance - estimation of variance components using double hierarchical generalized linear models*. Genetics Selection Evolution 42:8.
- Felleki, Lee, Lee, Gilmour and Rönnegård (2012) *Estimation of breeding values for mean and dispersion, their variance and correlation using double hierarchical generalized linear models*. Genetics Research 94:307-317.
- Rönnegård, Felleki, Fikse, Mulder and Strandberg (2013) *Variance component and breeding value estimation for genetic heterogeneity of residual variance in Swedish Holstein dairy cattle*. Journal of Dairy Science 96:2627-2636.

Linear mixed model using pedigree information

Animal model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{a} + \mathbf{e}$$

$$\mathbf{a} \sim N(\mathbf{0}, \mathbf{A}\sigma_a^2)$$

$$\mathbf{e} \sim N(\mathbf{0}, \sigma_e^2)$$

a_i = additive genetic effect for individual i

$\mathbf{A}$ = relationship matrix (calculated from pedigree information)

Extending the animal model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{a} + \mathbf{e}$$

$$\mathbf{a} \sim N(\mathbf{0}, \mathbf{A}\sigma_a^2)$$

$$\mathbf{e}_i \sim N(\mathbf{0}, \sigma_{e,i}^2), \log(\sigma_e^2) = \mathbf{X}_d\boldsymbol{\beta}_d + \mathbf{Z}_d\mathbf{a}_d$$

$$\mathbf{a}_d \sim N(\mathbf{0}, \mathbf{A}\sigma_{a_d}^2)$$

$$\boldsymbol{\rho} = \text{cov}(\mathbf{a}, \mathbf{a}_d)$$

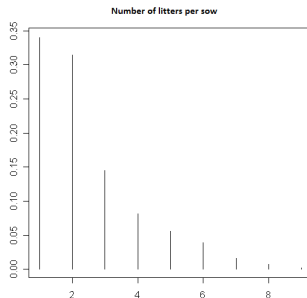
Sorensen and Waagepetersen (2003) developed a Bayesian method.

DHGLM possible to fit using existing variance-component estimation software (eg ASReml).

Data Description

Data from Danish Pig Production analyzed in Sorensen and Waagepetersen (2003).

- Pig litter size from 4,149 sows
 - mean litter size 10.3
- The data includes 10,060 records from these 4,149 sows in 82 farms.
- Fixed effects: farm, season, type of insemination, parity



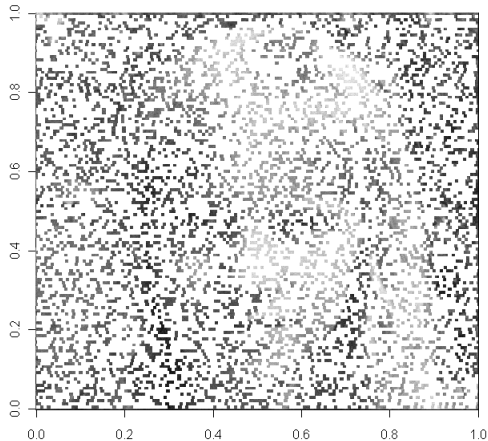
Simulation results

Table 3: Mean (a) and standard errors (b) of estimates of parameters for simulated data over Model III (first section) and IV (second section). The left hand column contains the simulated data structure.

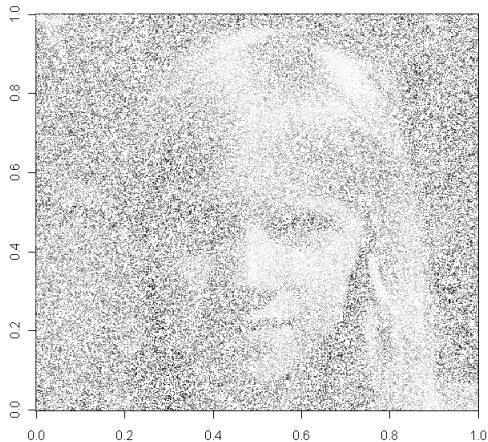
Column contains the simulated data structure.								
(a)	Mean model		Residual variance model					Cor
	Variances		Fixed effects			Variances		
	σ_a^2	σ_p^2	β_{d0}	β_{dins}	β_{dpar}	$\sigma_{a,d}^2$	$\sigma_{p,d}^2$	ρ
True values (Model III)	1.62	0.60	1.77	-0.17	0.35	0.09		-0.62
Original parity distribution	1.58	0.45	1.74	-0.17	0.33	0.11		-0.52
Four parities	1.59	0.61	1.73	-0.17	0.34	0.12		-0.56
Nine parities	1.63	0.60	1.74	-0.17	0.35	0.11		-0.60
True values (Model IV)	1.62	0.60	1.77	-0.17	0.35	0.09	0.06	-0.62
Original parity distribution*	1.56	0.24	1.73	-0.16	0.31	0.08	0.13	-0.61
Four parities	1.65	0.51	1.66	-0.16	0.34	0.09	0.15	-0.64
Nine parities	1.62	0.60	1.71	-0.17	0.35	0.09	0.09	-0.64
(b)								
(Model III)								
Original parity distribution	0.018	0.016	0.007	0.003	0.004	0.002		0.008
Four parities	0.015	0.012	0.005	0.003	0.004	0.002		0.007
Nine parities	0.012	0.008	0.005	0.001	0.003	0.001		0.005
(Model IV)								
Original parity distribution*	0.017	0.016	0.006	0.004	0.004	0.003	0.004	0.012
Four parities	0.017	0.012	0.007	0.003	0.004	0.002	0.003	0.008
Nine parities	0.013	0.008	0.005	0.002	0.003	0.001	0.001	0.005

* 27 out of 100 replicates did not converge. Estimates are for all replicates (with minor differences in results if these 27 replicates were included or not).

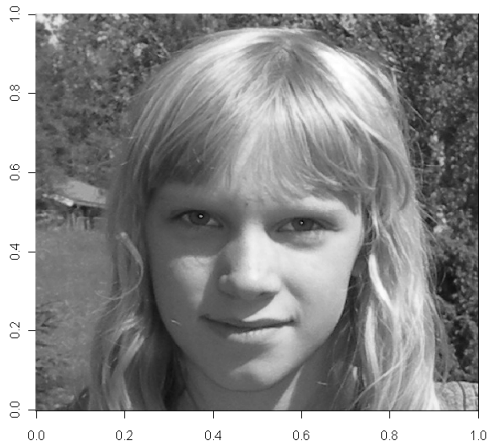
Image reconstruction



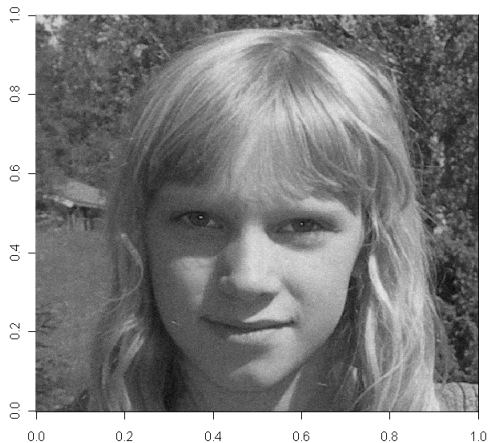
A last illustrative example - Image reconstruction



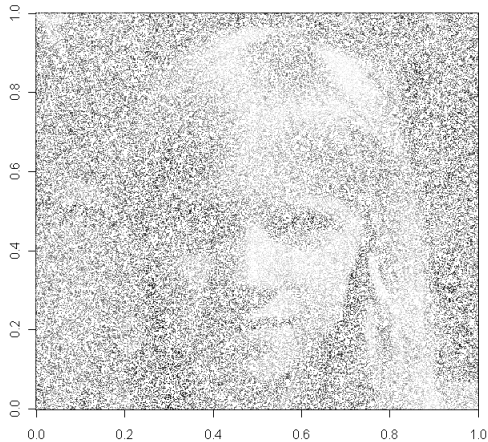
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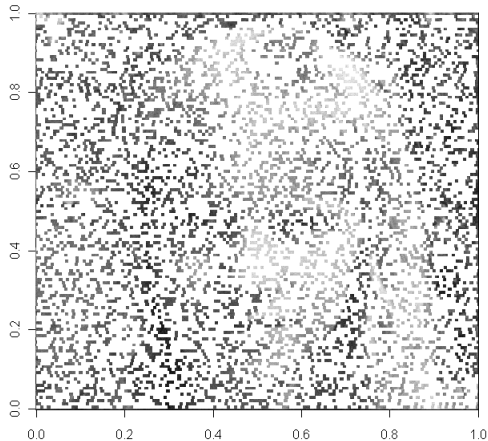
Noise added



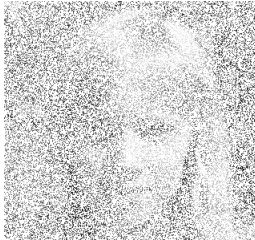
70% of pixels missing at random



Clustered 4x4 pixels missing



(a) 70 % pixels are missing at random



(c) Reconstruction of (a)



(b) 4×4 clusters are missing



(d) Reconstruction of (b)



Contact and Collaborators

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